

8DM60: Multimodal AI in Biomedicine

Integration of AI in the clinical workflow — Lecture 5

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Where we are in the course

Lectures 1–4 asked: *can a model do the task?*

Today asks: *can it be used?*

The message of this lecture

Most clinical AI does not fail at **Q4 (architecture)** or **Q5 (objective)**.

It fails at **Q8 (how would it be used in practice?)** and **Q9 (what could go wrong?)**.

1. Task
2. Data
3. Preprocessing
4. Architecture
5. Objective
6. Training / adaptation
7. Evaluation
- 8. Use in practice**
- 9. Risks and mitigation**

Today: the workflow as a system → why integration is hard → integration patterns.

Outline

1. The clinical workflow as a system

Where does the model actually connect?

2. Why integration is the hard part

The gap between published models and deployed products

3. Integration patterns and where the model sits

Triage, assistant, second reader, autonomous

4. Validation for deployment vs. validation for publication

distinguish between models for research and clinical deployment

5. Human factors and the sociotechnical view

bias and explainability

6. Regulatory, safety, and governance

Regulation, liability, and ethics; effects of model deployment

7. Economics and sustainability

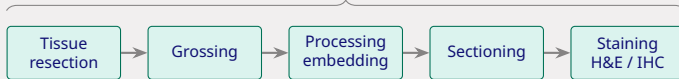
reimbursement, cost-effectiveness, and business models

Part 1

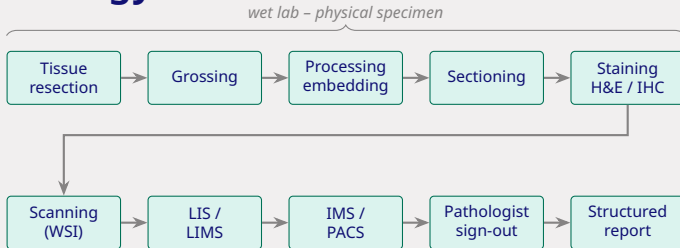
The clinical workflow as a system

The histopathology workflow: tissue to treatment

wet lab – physical specimen

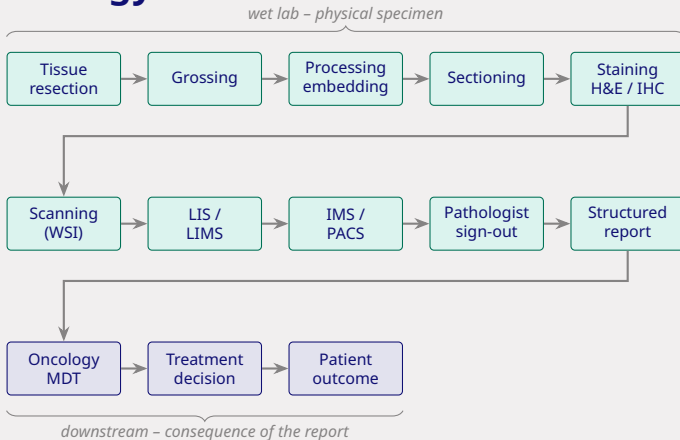


The histopathology workflow: tissue to treatment



Note the asymmetry: the first five steps are **physical**, the next five are **informational**. AI can only enter after digitization, which is why scanner deployment is the rate-limiting step in most labs.

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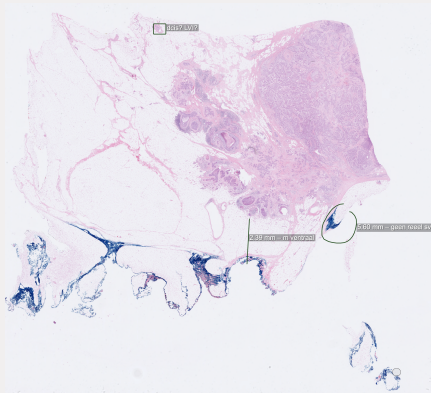
Whole-slide images: what digitisation produces

One slide

- Scanned at $40\times$ ($\approx 0.25\text{ }\mu\text{m}/\text{pixel}$)
- $\approx 1\text{ GB}$ per slide on average

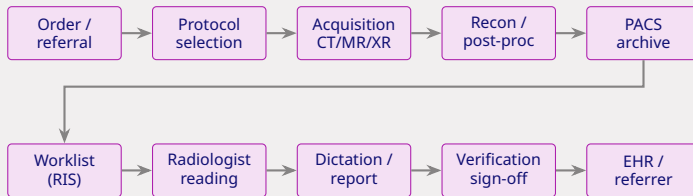
One medium-sized Dutch lab, per year

- **40,000** cases
- **250,000** slides (≈ 6 per case)
- $\approx 250\text{ TB}$ of new image data
- $\approx 1,000$ slides, $\approx 1\text{ TB}$ *per working day*



Figures for a medium-sized Dutch pathology lab; storage assumes $\approx 1\text{ GB}$ per slide at $40\times$.

The radiology workflow: the same shape, one decade earlier



Structural equivalences

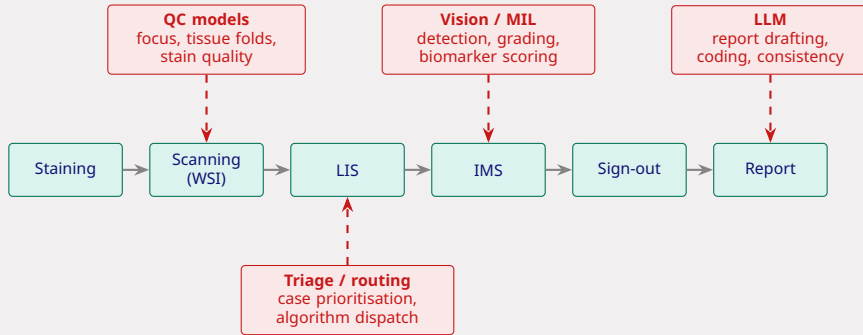
Scanner ↔ modality
IMS ↔ PACS
LIS ↔ RIS
Slide tray ↔ worklist
Sign-out ↔ verification

Why radiology is ahead

Born digital — no scanning step
DICOM mature since the 1990s
Files ~MB, not ~GB
Established AI reimbursement codes
IHE profiles for AI results

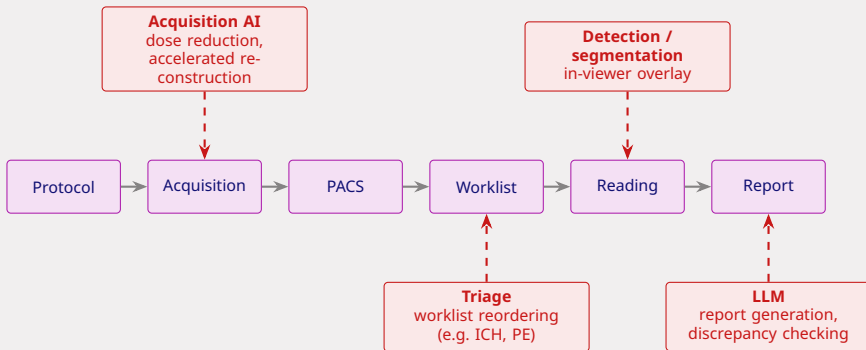
The pattern is reusable. Pathology is repeating radiology's integration history under harder data constraints.

Where AI plugs into the pathology workflow



Four *different* kinds of model, four *different* integration points, four *different* consequence-of-error profiles. A QC model that misfires costs a rescan. A grading model that misfires costs a treatment decision.

Where AI plugs into the radiology workflow



Radiology AI touches the imaging chain at ordering, preprocessing, acquisition, postprocessing, reporting and storage — the degree of required human oversight differs at every one of these points.

Kohli et al., *Radiology*, 2024 — AI touch points and IHE integration profiles.

Commercial systems in routine use

Radiology

System	Task	Integration point	Evidence
Transpara (ScreenPoint)	Mammography detection + triage	Screen-reading workflow, replaces one reader	MASAI RCT, $n > 105,000$
Lunit INSIGHT MMG	Mammography detection	Assistant reader in double-read pathway	Prospective non-inferiority + health-economic evaluation
Gleamer BoneView	Fracture / effusion detection on X-ray	Native PACS/RIS orchestration layer	Deployment case study (Vestre Viken, NO)
LumineticsCore (ex IDx-DR)	Diabetic retinopathy	<i>Autonomous</i> , point of care	FDA De Novo; CPT 92229 reimbursement

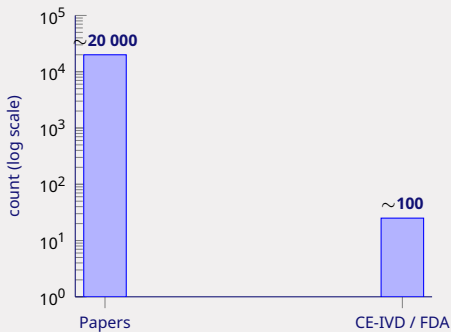
Pathology

System	Task	Integration point	Status
Paige Prostate Detect	Cancer detection on prostate biopsy	In-viewer, adjunct at sign-out	First FDA-authorised AI pathology product (De Novo, 2021); CE/IVDR CE-IVD modules, widely used in EU labs
Visiopharm	IHC quantification, biomarker scoring	Image analysis in IMS	
Lunit SCOPE PD-L1	PD-L1 TPS scoring (NSCLC)	Embedded in HALO AP, Aperio, navify	CE-marked; RUO in the US
Ibex Galen	Prostate / breast QC and detection	Pre-sign-out screening	Deployed in EU / IL / US laboratories

Part 2

Why integration is the hard part

Publications versus products



Computational pathology, order-of-magnitude estimate

The adoption paradox

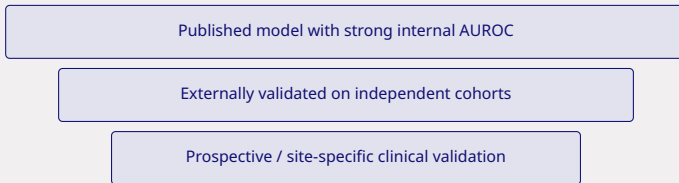
Foundation and multimodal models now match expert-level performance on many diagnostic, prognostic and biomarker tasks.

Yet only a handful of AI systems are used in routine clinical practice — and the products that reached patients differ substantially from research prototypes.

Three orders of magnitude.

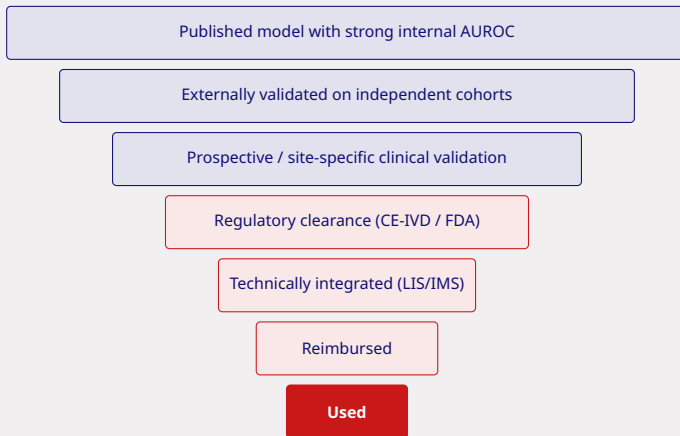
Cai et al., *LabMed Discovery* 3(2):100130, 2026.

Why the funnel is so narrow



Steps 1–3 are what papers report.

Why the funnel is so narrow



Steps 1–3 are what papers report. Steps 4–7 are where systems die, and they are almost never addressed in the paper that introduced the model.

Model scope versus clinical case mix

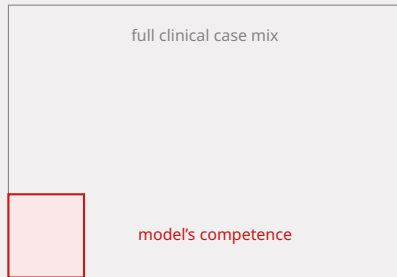
The published model

“Classification of melanocytic lesions”

benign naevus vs. melanoma, curated WSIs, balanced classes, single scanner

The dermatopathology bench

basal cell carcinoma, squamous cell carcinoma, actinic keratosis, seborrhoeic keratosis, inflammatory dermatoses, cysts, adnexal tumours, re-excisions, *and* melanocytic lesions — plus the ambiguous, the unstained, the poorly oriented and the artefactual



Schematic, not to scale

Q for discussion: what does the pathologist do with the *other* slides? If the answer is “look at all of them anyway”, the model has not saved any time.

The generalisation problem is not a data-quality problem

Developer's claim

0.76–0.83

AUC in vendor internal
documentation for a proprietary
sepsis prediction model already
running in hundreds of US hospitals

Wong et al., *JAMA Intern Med* 2021;181(8):1065–1070.

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External validation

0.63

AUC in an independent study: 38 455 hospitalisations, single centre

What that meant at the bedside

- Sensitivity 33%, PPV 12%
- Missed sepsis in **67%** of septic patients
- Alerted on 18% of *all* patients
- Caught only **7%** of the septic patients clinicians had themselves missed

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- Missed sepsis in **67%** of septic patients
- Alerted on 18% of *all* patients
- Caught only **7%** of the septic patients clinicians had themselves missed

You cannot detect this with a better test set drawn from the same source. The model added least value exactly where clinicians already needed it most.

Wong et al., *JAMA Intern Med* 2021;181(8):1065–1070.

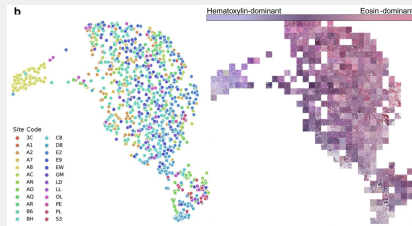
The same failure in histopathology: site signatures

What was shown

- Histologic image differences between *submitting sites* in TCGA are easily identified by a deep-learning model
- These signatures vary substantially across sites, over 3 000 patients and six cancer types
- Submitting site correlates with outcome base rates — so a model can score well by learning **where the slide came from**

Why internal validation misses it

Site is present in both the training and the test folds. A random split does not separate the confounder from the label, so cross-validation reports inflated accuracy and reports it *confidently*.



The model learned a shortcut that was **real, reproducible and predictive** — inside the training distribution.

It carries no biology, so it does not transfer.

Howard et al., *Nat Commun* 2021;12:4423.

Model output versus clinical need

What the model outputs	What the report needs
Slide-level probability of malignancy	Tumour type, grade, extent, margins, stage
Heatmap over a WSI	A sentence a clinician can act on
A number, per slide	A conclusion, per <i>patient</i> , across all blocks
Binary label at a fixed threshold	A decision under an explicit clinical policy

Recall the FDA's own caveat on the first authorised AI pathology product: the pivotal study *did not evaluate the impact on final patient diagnosis, which is typically based on multiple biopsies*.

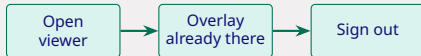
The unit of the model is the slide. The unit of care is the patient.

Context switching: the productivity killer

Not integrated



Integrated



Every context switch is a transcription opportunity, an interruption, and a reason not to use the tool. A system that adds clicks will be abandoned regardless of its AUROC — professional guidance is explicit that AI must not add extra burden to the reader.

Integration is a standards problem, not an ML problem

The plumbing

- **DICOM** — images, including WSI
- **HL7 v2 / FHIR** — orders and results
- **SNOMED CT** — coded findings
- **IHE profiles** — how the above are used *together*

In the IHE role model the AI algorithm is simply an “*evidence creator*” — one component in a pre-existing architecture it does not get to redefine.

The proprietary trap

Some implementations store AI results in proprietary formats or private DICOM attributes, requiring proprietary viewers and unrealistic customisation burdens.

Multiply by n vendors $\times$ m algorithms.

Abels & Pantanowitz et al., *J Pathol Inform*, 2021; Kohli et al., *Radiology*, 2024.

Who pays? Four different business cases

System	Customer	Value proposition	Measured in
Paige Prostate	Pathology lab	Fewer IHC orders, faster turnaround	Lab cost, TAT
Transpara / Lunit MMG	Screening programme	Replace one of two readers	Cost per cancer detected
LumineticsCore	Primary care practice	Billable point-of-care test	CPT 92229 fee
Lunit SCOPE PD-L1	Pharma / CRO	Trial enrichment, stratification	Drug development risk

The buyer determines the endpoint. A tool sold to a lab must show efficiency; a tool sold to a screening programme must show outcomes; a tool sold to pharma need never show either.

Reimbursement does not equal adoption

Autonomous diabetic retinopathy screening has what every other clinical AI wants:

- FDA De Novo authorisation (2018)
- A dedicated billing code, CPT 92229
- A fee *above* the human equivalent: \$40.28 vs. \$29.14 (MD read), \$17.35 (staff review)

And yet

Across a database of >107 million patients, **under 5%** of diabetic patients received *any* ophthalmic imaging for retinopathy (2019–2023) — and only a small fraction of those were AI-based.

Regulatory approval, integration and payment are *necessary*. They are not *sufficient*.

JAMA Ophthalmology cohort study of CPT 92229 utilisation; reimbursement rates from 2023 fee schedules.

Part 3

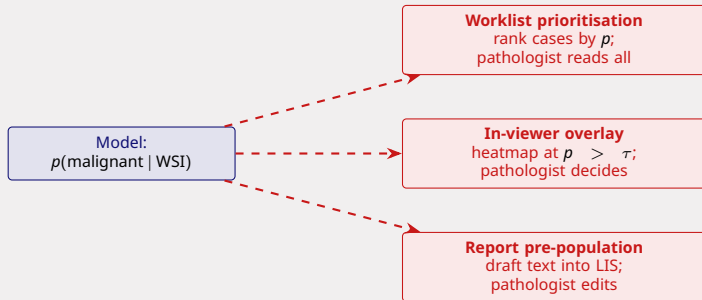
Integration patterns and where the model sits

Four integration patterns

Pattern	Model acts	Failure mode	Example
Triage / pre-screen	Before the human, reorders the queue	Delay, not error — a missed case is still read	Worklist prioritisation for intracranial haemorrhage
Concurrent assistant	Alongside the human, in the viewer	Automation bias; anchoring	Paige Prostate; Transpara detection support
Second reader / QC	After the human, flags discrepancy	Alert fatigue; over-calling	Discrepancy checking on reports; slide QC
Autonomous	Instead of the human	Undetected error reaches the patient	LumineticsCore for retinopathy

Required validation depth increases down the table. So does regulatory burden, and so does liability exposure.

The same model, three different products



Identical model. Completely different implementation — different users, different failure modes, different evidence requirements, and in the EU, potentially different device classifications.

Human-in-the-loop vs. human-on-the-loop

Human *in* the loop

Every output passes through a human before it acts.

Risk: automation bias — the human stops being an independent check and starts confirming the machine.

Requires: the human to retain competence at the unassisted task.

The uncomfortable question: if the AI is right 99% of the time, what exactly is the human contributing — and are they still able to contribute it after a year of agreeing?

Human *on* the loop

The system acts; the human monitors and can intervene.

Risk: vigilance decrement — monitoring a reliable system is a task humans are bad at.

Requires: drift monitoring, audit trails, and a defined escalation path.

Liability follows the pattern

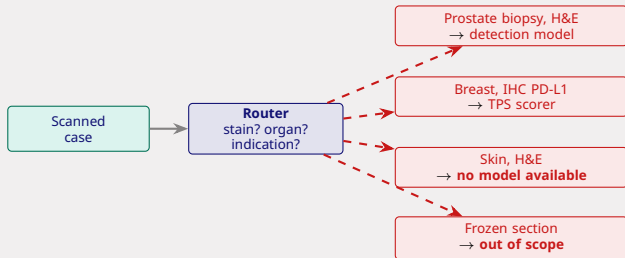
Pattern	Who is accountable	Open question
Assistant / second reader	The signing pathologist or radiologist	Is “the AI missed it” ever a defence?
Autonomous	Arguably the manufacturer	The AMA has argued the creator or vendor should assume medical liability

Autonomous systems are typically deployed *where the specialist is absent* — e.g. a primary care physician running a retinal screen. That physician has no specialist knowledge of the retina and arguably should not carry the liability for an incorrect result.

Q9 is not only technical.

The routing problem — pathology and radiology

Pathology routing



Radiology routing

Dispatch on modality + body part + clinical indication, read from the DICOM header and the RIS order.

Harder than it sounds: protocol names are institution-specific, headers are inconsistent, and one study may need three algorithms or none.

This is why IHE defines profiles for AI workflow and worklist prioritisation — routing is a *standardised* problem, not a per-hospital script.

Who builds the router? It is not part of any vendor's product, it needs LIS/RIS metadata that may be free text, and it must fail *safely*: a mis-routed case analysed by the wrong model is worse than no model at all.

Part 3: what to take away

1. The workflow is the system; the model is one component.
2. Radiology and pathology share a structure — reuse the pattern.
3. Q8 and Q9 kill more systems than Q4.
4. The integration pattern determines the required evidence, the failure mode and the liability.
5. Ask “who pays, and for which endpoint?” before “what is the AUROC?”

Part 4

Validation for deployment vs. validation for publication

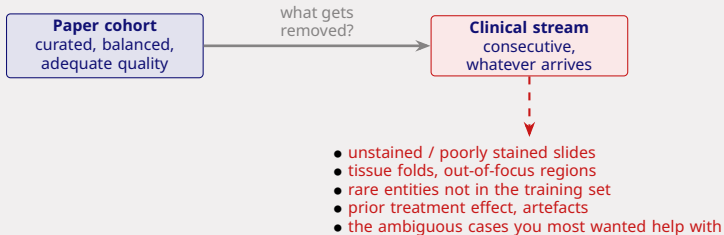
Four kinds of validation, three of which papers skip

Type	Question answered	Data	Typical venue
Internal	Did the model learn the training distribution?	Held-out split, same source	Every paper
External	Does it transfer to another centre?	Independent cohort, different scanner/lab	Good papers
Prospective	Does it work on tomorrow's cases, read in order?	Consecutive, unselected, real time	Rare
Site-specific clinical	Does it work <i>here</i> , in <i>our</i> lab, before go-live?	Local cases on local equipment	Almost never published

Professional guidance for whole-slide imaging is explicit: validation should emulate the laboratory's actual clinical environment, using at least 60 routine cases per application.

CAP/ASCP/API guideline update, *Arch Pathol Lab Med* 146(4):440–450, 2022.

Retrospective curated $\neq$ prospective consecutive



How deployed models degrade

Mechanism	Cause in pathology / radiology	Detectable by
Covariate shift	New scanner, new stainer, changed H&E protocol, new reagent lot	Monitoring input statistics (stain vectors, focus scores)
Label / prior shift	Case mix changes; new referral pattern; screening interval changes	Monitoring output distribution vs. baseline
Concept drift	Guideline revision changes what counts as the target class	Only by human audit
Feedback loop	Model influences the labels it is later retrained on	Very hard — see next slides

The silent failure: none of these produce an error message. The model keeps returning confident, well-formatted, plausible output.

Monitoring a model you cannot evaluate

The core difficulty

Ground truth in production arrives **late, partially,** or **never**:

- Interval cancers surface after 2 years
- Benign cases get no follow-up
- Discrepancies are only found if someone re-reads

So you monitor *proxies* and audit *samples*.

A minimum monitoring plan

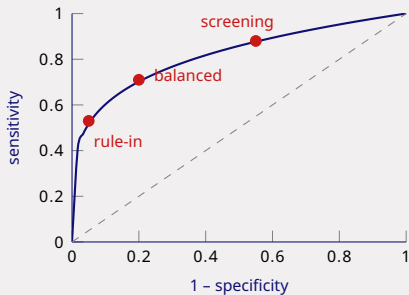
1. Input drift metrics per scanner and per stainer
2. Output rate vs. historical baseline, by subgroup
3. Rate of human override / disagreement
4. Periodic blinded re-read of a random sample
5. A defined threshold that triggers withdrawal

Guidance requires procedures for changes to the imaging system that could affect clinical results — the same must hold for the model.

Metrics: what papers report vs. what deployment needs

Published	Deployment	Why the difference matters
AUROC (threshold-free)	Sensitivity/specificity <i>at the chosen operating point</i>	You must ship one threshold. AUROC does not tell you which
Balanced accuracy on balanced data	PPV / NPV at the true prevalence	At 1% prevalence a 95%-specific test is mostly false positives
Slide-level metrics	Patient-level, case-level outcomes	A patient has multiple blocks and one diagnosis
Mean performance	Worst-subgroup performance	Physicians need to know on which subpopulations models fail
Accuracy	Interval cancer rate, turnaround time, override rate	The endpoint the payer and the regulator care about

Choosing the operating point is a clinical policy decision



Schematic ROC — one curve, three products

One model, one AUROC, three different clinical tools.

Screening / pre-screen: a sensitivity floor is non-negotiable; false positives are absorbed by the human.

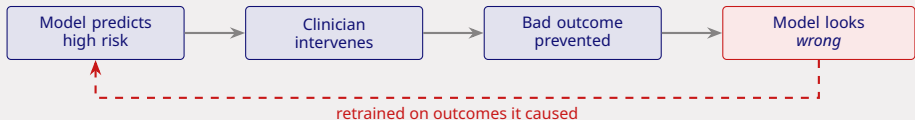
Autonomous rule-out: NPV must be near-perfect at the local prevalence.

The threshold encodes a value judgement about whose error matters more. That judgement is clinical, not technical — but it is almost always made by the developer.

The prediction paradox

The problem

A prediction model that is *acted upon* destroys the conditions under which it was validated. The better the response to the prediction, the worse the prediction appears — or the more it appears to confirm itself.



The mirror image is the **self-fulfilling prophecy**: a model predicts a poor outcome, care is de-escalated, the poor outcome occurs, and the model is “validated”.

Why the paradox matters for pathology AI

Where it bites

- Any model whose output changes what is *ordered next* (extra levels, IHC, molecular testing)
- Any model retrained on data generated after its own deployment
- Triage models: cases the model deprioritised may be read later, less carefully, by a more tired reader

Mitigations

- Randomised deployment (as in MASAI) — the only clean answer
- Hold out a non-AI arm or period
- Record the model output *and* the action taken separately
- Never retrain naively on post-deployment labels

Framework link: this is a Q7 problem that only becomes visible when you take Q8 seriously. You cannot detect it with a better test set.

Part 5

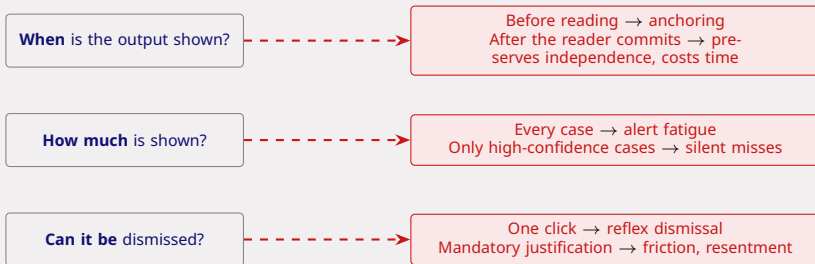
Human factors and the sociotechnical view

The bias catalogue

Bias	What happens	Consequence
Automation bias	The reader accepts the AI output without independent assessment	AI errors become human errors
Automation complacency	Reduced vigilance because the system is usually right	Misses in the cases the AI did not flag
Anchoring	The overlay is seen before the reader forms an opinion	The independent second opinion is lost
Alert fatigue	Too many low-value flags	All flags dismissed, including the true ones
Deskilling	Unassisted competence decays over time	No safe fallback when the system is down
Rejection bias	Distrust after a salient early error	The tool is bypassed entirely

Note that these pull in *opposite* directions: fixing over-trust can create under-trust. The target is **calibrated** trust, not maximal trust.

Design choices that determine which bias you get



There is no neutral choice here. **The user interface is part of the medical device** — and in the EU, legally so.

Explainability: probability is not confidence

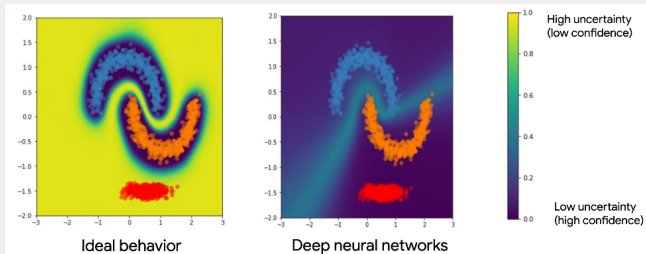
What the model gives you
A softmax output of **0.94**.

What the reader assumes
“It is 94% sure.”

Two very different things are being
conflated:

Calibration — of all cases scored 0.94,
are 94% positive?

Epistemic uncertainty — is this case
even in distribution?



A model is typically confident and wrong precisely on out-of-distribution input — the rare entity, the unusual stain, the new scanner. That is exactly where the reader most needs the model to say “I don’t know”.

Figure: Liu et al., “Simple and principled uncertainty estimation with deterministic deep learning via distance awareness”, NeurIPS 2020.

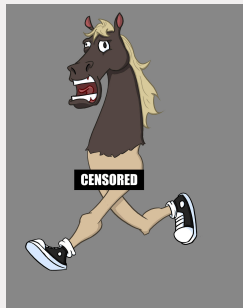
Explainability where it actually matters

Method	What it shows	What the user needs
Attention / saliency heatmap	<i>Where</i> the model looked	<i>Why</i> that region is diagnostic
Feature attribution	Which inputs moved the score	Whether the reasoning matches morphology
Confidence score	A number	Whether to trust <i>this</i> case

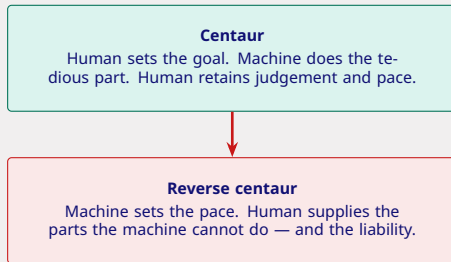
What actually helps at sign-out: the ability to jump to the region and look at it. A heatmap that points at a real focus of tumour is useful because the pathologist verifies it — not because the heatmap explains anything.

Reframe: the useful question is not “is the model explainable?” but “*can the user efficiently verify or refute this output within their existing workflow?*”

Centaur and reverse centaur



Reverse Centaur by [m-watts-art](#) (DeviantArt)



The term is Cory Doctorow's. The distinction is not about the technology — it is about **who is in service of whom**. The same detection model can produce either arrangement depending on how the workflow around it is designed.

which arrangement does a productivity-target-driven deployment tend to produce?

Symptoms of a reverse-centaur deployment

Warning signs

- Throughput targets set on the assumption the AI is right
- The reader must justify *disagreeing*, not agreeing
- Override rate is monitored as a *problem* to be reduced
- The tool cannot be switched off for a difficult case
- Time saved is reinvested as more cases, not more care per case

The honest tension

The business case for most clinical AI is **throughput**.

The safety case requires **unhurried human judgement**.

These are in direct conflict, and the conflict is resolved by management decisions, not by the model.

Change management matters as much as the algorithm: implementing AI involves complex social and technical processes requiring significant resources — training, engagement, governance and impact monitoring.

NHS rapid evaluation of AI for chest diagnostics, 2025.

Part 6

Regulatory, safety and governance

The European regulatory stack, in headlines

Instrument	What it governs	Relevance here
MDR 2017/745	Medical devices, incl. standalone software	Radiology AI; class rises with decision risk
IVDR 2017/746	In vitro diagnostics	Pathology AI on specimens; Paige Prostate holds IVDR approval
EU AI Act	AI systems by risk tier	Medical-device AI is high-risk; adds transparency, data-governance, human-oversight duties
GDPR	Personal data	Training data provenance, secondary use
EHDS	Health data spaces	Secondary use for AI development

Note the layering: an IVDR-certified model is *also* a high-risk AI system. Compliance with one does not discharge the other.

Headline orientation only — verify current guidance and transition timelines before relying on this.

Locked vs. adaptive, and the hospital's own duties

Locked vs. adaptive

Once cleared, model parameters cannot be changed without going back to the regulator. In practice almost all deployed clinical AI is **locked**.

This is why the drift problem in Part 4 is acute: the world moves and the model cannot.

Predetermined change-control plans are the emerging answer; treat as an evolving area.

What the hospital must do anyway

CE marking is the manufacturer's claim. The *deploying* lab still owns:

- Site-specific validation before go-live
- Intended-use compliance (off-label use voids it)
- Incident reporting and vigilance
- Audit trail: which model version, on which case, when
- Decommissioning plan

This sits inside the existing quality system (ISO 15189), not beside it.

Liability: the pathologist signs the report

In the **assistant** and **second-reader** patterns, the accountable party is unchanged: the professional who signs.

The device is authorised as an *adjunct*. Intended use says the professional evaluates the case, taking account of patient history — the AI output is one input among several.

So: is “the AI missed it” a defence?

And symmetrically — is “the AI flagged it and I disagreed” a liability *increase* for the pathologist who was right to disagree?

The autonomous case is different
Autonomous systems are deployed where the specialist is absent. The operating clinician has no specialist knowledge of the domain and arguably should not carry liability for an incorrect result — the AMA has argued the developer or vendor should assume it.

No settled European position; watch the AI Liability discussions.

A practical governance checklist

Before go-live

1. Who owns this model in the organisation?
2. Intended use, written down, and how we will stay inside it
3. Site-specific validation report, signed
4. Which version is running, and how do we know?
5. What is the fallback when it is unavailable?

After go-live

1. Monitoring plan with named thresholds
2. Route for staff to report suspected AI errors
3. Periodic re-validation cadence
4. Subgroup performance review
5. Criteria and authority to **switch it off**

Most organisations have a procurement process for AI and no decommissioning process. The second is the one that protects patients.

Equity and bias as a governance problem

Where disparity enters

- Training cohorts from a few academic centres in a few countries
- Acquisition hardware and protocols — scanners, stains, imaging devices — correlate with the resources of the site
- Patient and specimen factors shift the input: body habitus, positioning, tissue handling
- Reference standards derived from the same non-representative population
- Rare entities and small subgroups under-represented by construction
- Unequal access to imaging or digitisation at all

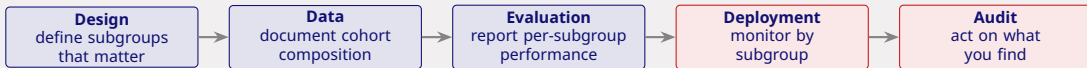
Recall also the access gap: a tool can be accurate, approved, integrated *and* reimbursed, and still reach under 5% of the eligible population.

Why this is governance, not just science

A mean performance figure can hide a subgroup for which the model is useless or harmful.

Nobody detects this unless **someone is required to look** — so it belongs in the monitoring plan, with a named owner and a threshold, not in the discussion section of a paper.

Bias mitigation is a lifecycle activity



The step that is almost always missing is the last one. Detecting a disparity and having no mechanism, budget or authority to respond to it is worse than not measuring — it converts an unknown risk into a documented, unaddressed one.

Part 7

Economics and sustainability

Return on investment: what is actually in the equation

Benefit side (often overstated)

- Reader time released — only banked if it is *redeployed*
- Fewer ancillary tests (IHC, molecular)
- Shorter turnaround time
- Errors avoided — real, but hard to price
- Capacity gained without hiring

Cost side (usually understated)

- Licence fee (per case? per seat? per year?)
- **The scanner and the digital pathway itself**
- Storage and compute at scale
- Integration engineering and IMS/LIS licences
- Validation before go-live, and re-validation
- Monitoring, governance and staff training
- Downtime and fallback processes

The trap: time saved per case is real but *fractional and distributed*. Fifteen seconds per slide across a department does not become a post you can cut — and that is what a business case usually has to demonstrate.

Who pays, when, and how

Mechanism	How it works	Example / status
Dedicated code	A billing code exists for the AI act itself	US CPT 92229 for autonomous retinal analysis, \$40.28 (2023)
Bundled	AI is absorbed into the existing tariff for the test	The default in most European systems — the lab pays, nobody reimburses
Programme budget	A screening programme buys capacity	Mammography AI replacing a reader
Central / national fund	Time-limited award to seed adoption	NHS AI in Health and Care Award
Industry pays	Pharma or CRO buys the readout	Lunit SCOPE PD-L1 for trial stratification

Europe's structural problem: in a bundled DRG or tariff system, the department that pays for the AI is often not the one that captures the saving.

Reimbursement does not equal adoption — again

The retinopathy case is the cleanest natural experiment available:

- Regulatory authorisation: **yes** (2018)
- Integration into the EHR: **yes**
- Dedicated payment code: **yes** (2021)
- Paid *above* the human alternative: **yes**
- Adopted at scale: **no**

For your business case: budget for the implementation programme, not just the licence. In published deployments the implementation effort routinely dwarfs the software cost.

What was still missing

Stakeholder alignment across primary care and specialty, workflow adaptation, site selection, staff ownership, and performance tracking.

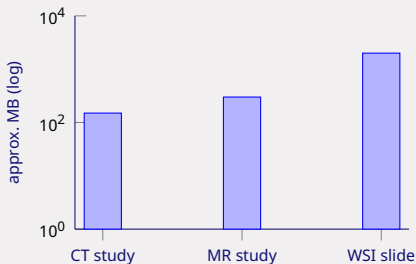
None of these are properties of the model.

Who is the customer for the model output?

	Paige Prostate	Lunit SCOPE PD-L1	Transpara / MASAI
Buyer	Pathology laboratory	Pharma / CRO	Screening programme
User	Pathologist at sign-out	Trial operations	Screening radiologist
Value	Accuracy, fewer IHC, faster TAT	Patient stratification, trial enrichment	Reader capacity, detection rate
Endpoint	Lab cost and turnaround	Drug development risk	Interval cancer rate
Regulatory	FDA De Novo, IVDR	CE-marked; RUO in the US	CE-marked

The instructive comparison: Lunit sells a mammography product to hospitals *and* a PD-L1 product to pharma. Same company, same underlying competence, two completely different business-case architectures — and only one of them needs a hospital ROI calculation at all.

What it costs to run: the storage problem



Order of magnitude only — verify against your own archive

A mid-sized pathology department producing hundreds of thousands of slides a year moves into **petabyte** territory within a few years of going fully digital.

And the retention obligation is measured in *decades*, not years.

The AI is not what makes digital pathology expensive. Digitisation is. The AI is the justification offered for an infrastructure cost that has to be paid first.

The maintenance burden nobody budgets for

Recurring, for the life of the deployment

- Re-validation after *any* change: scanner, stainer, protocol, IMS version, model version
- Monitoring and periodic audit
- Vendor management and version control
- Staff turnover → retraining
- Storage growth, indefinitely
- Incident investigation

Sustainability questions

- What happens when the vendor is acquired or exits?
- Can you export your data and annotations?
- Is the model still supported in five years?
- Who pays for the re-validation the vendor's update forces on you?

Ask these *during* procurement. They are unanswerable afterwards.

Pulling it together

1. The workflow is the system; the model is one replaceable component.
2. Radiology and pathology share a structure — the integration pattern is reusable.
3. Q8 and Q9 kill far more systems than Q4.
4. The integration pattern determines the evidence required, the failure mode, and who is liable.
5. Validation for deployment is a different activity from validation for publication.
6. Deployed models degrade silently; monitoring is not optional.
7. The interface is part of the device — it decides which human bias you get.
8. Approval, integration and payment are necessary and not sufficient.
9. Always ask: who is the customer, and for which endpoint?

One sentence

A model that is never used has an AUROC of 1.0 and a clinical utility of zero.

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